

Data Science

Course Introduction



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SoftUni



Software University

<https://softuni.bg/>

Have a Question?

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#DataScience

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Table of Contents

1. Course objectives
2. Prerequisites
3. Curriculum
4. Course schedule
5. Trainer
6. Lecture format
7. Final exam
8. Some learning resources





Course Objectives

- Learn how to apply the scientific method to
 - Ask the right questions
 - Obtain and clean up data
 - Explore and analyze data
 - Make the correct conclusions
- Write your own research
 - Learn how to create evidence-based, reproducible research
- Learn how to create a complete solution
 - Incorporate best practices in software design and science
- Communicate and compare results with the community



Programming Basics

- Some familiarity with Python is required
- Software development experience is a plus



Math Concepts

- Know some algebra and statistics (and a little bit of calculus)
- Have basic logic and intuition



Intermediate English

- Understand what is written on the slides



Scientific Mindset

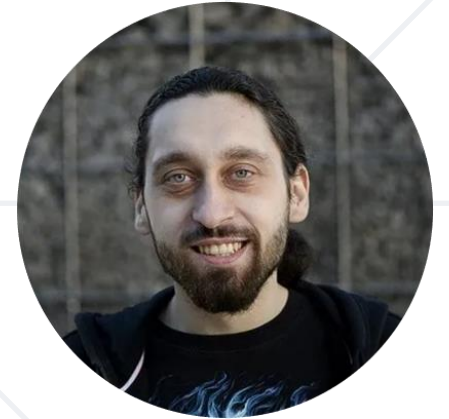
- Be open to (and not afraid of) challenges



Trainer

Who I am

- Trainer
- Software engineer
- Data Scientist / Machine Learning Engineer
- Scientist / Enthusiast



What I love

- Teaching (duh!)
- Nature
- Music



Course Organization

- Course Introduction
- Data Acquisition
- Data Tidying and Cleaning
- Data Visualization. Exploratory Data Analysis
- Working with Images
- Working with Text
- Regression Models
- Data Science Project Architecture
- Data Science in Production
- Exam Preparation and Q&A

24 June 2025

19 August 2025

23-24 August 2025

6-7 September 2025

Data Science
9 weeks * 1 time / week
9 credits

Start: 24 June 2025
End: 8 September 2025

Exam Sessions

Regular Exam: 23 and 24 August 2025
Retake Exam: 6 and 7 September 2025

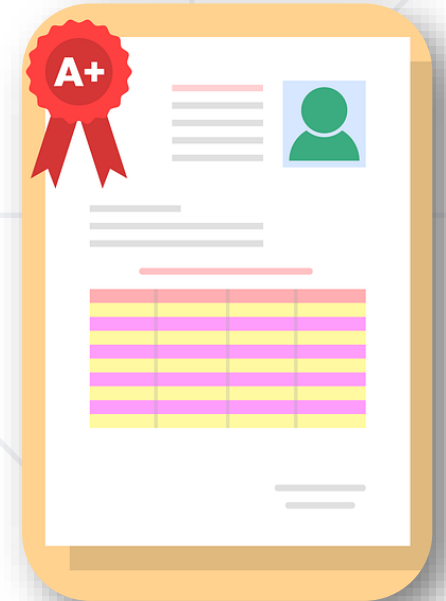
- **Lessons**
 - 9 lectures x 4 hours each
- **Homework**
 - 4 hours+ – the more, the better
 - Quiz – 0.25-0.5 hours
 - Questions to check your understanding
 - Lab – 1-6 hours
 - Problems related to real-life data science
- **Extracurricular activities: 0+ hours**
- **Practical exam**
 - 5-20+ hours



- **Quiz** (theoretical exam)
 - 10 questions for 30 minutes
- Practical project
 - Work on your own
 - No teams allowed
 - Present your results (documentation, code, etc.) in a **limited** amount of time
 - Work on a given assignment
 - Assignment release time: at second lecture
 - Perform research
 - Scientific papers, community forums, etc.
 - Analyze the data
 - Write code
 - Communicate the results

Grading Scheme

- **Quizzes:** up to 10%
 - Due date: at the end of the course
- **Labs:** up to 10%
 - Due date: at the end of the course
- **Final exam:** up to 80%
 - Theoretical exam (quiz): 30% (24% of total grade)
 - Practical exam (project): 70% (56% of total grade)
 - Develop at your own pace
 - **Upload deadline:** Friday before the exam date; 12:00 PM
 - Project defense
 - Online: Sunday, according to schedule
- **Discord / Facebook group activity:** bonus up to 10%
- **Other bonuses:** up to 10%



- All students will be graded on a **scale from 2.00 to 6.00**
 - The same way the standard grading in Bulgaria works
- Everyone who **scores ≥ 5.00** (total) will get a **certificate from SoftUni**
- Everyone who **scores ≥ 3.00** (on both theory and practice) can get a **MoE certificate** as well
 - You need to apply explicitly within a **limited time**



Why Bother?

- Starting point for a **new career** or **continuing education** in your current field
- **Career assistance**
 - The SoftUni career center will help you find work
- Official and recognizable
 - Employers value certificates
- Proof of hard work :)
 - Shareable and verifiable
- We make sure that everyone who scores more than 5.00 knows what they're doing :)

Course, Facebook Group and Discord Group

- Module Page:

<https://softuni.bg/modules/148/ai-and-machine-learning-march-2025/1491>

- Course Page:

<https://softuni.bg/trainings/4652/data-science-june-2025>

- Official Facebook group:

<https://www.facebook.com/groups/datasciencejune2025>

- Discord Channel:

<https://discord.gg/akJ9Z39Xq6>



■ Books

- ["How not to be wrong"](#) – Jordan Ellenberg
- ["Python Data Science Handbook"](#) – Jake VanderPlas
- ["Python for Data Analysis"](#) – Wes McKinney
- ... and anything else you can find

■ Websites

- [Khan Academy](#)
- Communities: [Kaggle](#), [Quora](#), [Stack Exchange](#)
- Online courses: [Coursera](#), [edX](#), [MIT OCW](#), [Stanford](#), etc.

■ YouTube

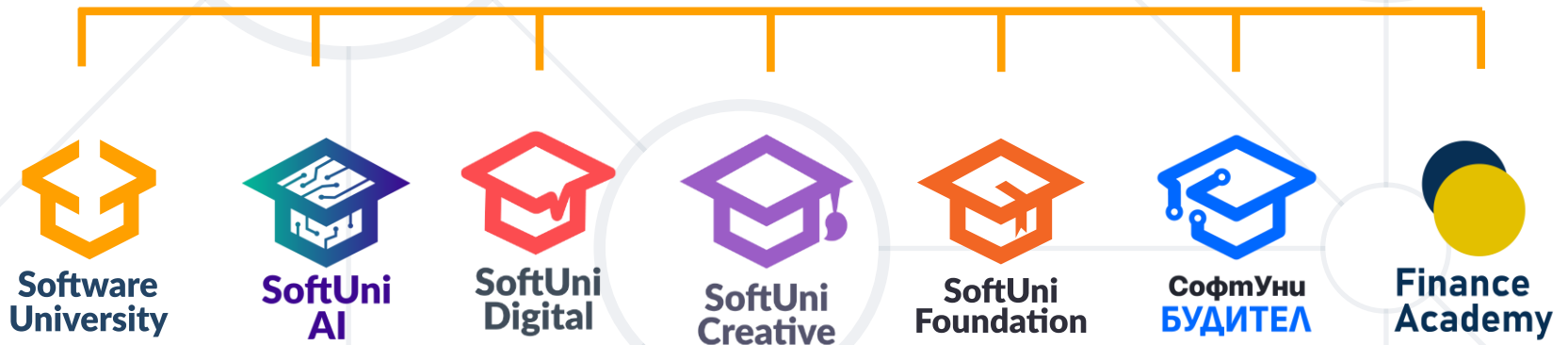
- [AsapSCIENCE](#), [Veritasium](#), [Vsauce](#), [TedEd](#), [Daniel Shiffman](#), [CrashCourse](#), [Numberphile](#), [Computerphile](#), [Vi Hart](#), [3Blue1Brown](#), [blackpenredpen](#), [Mathologer](#), and many more



Questions?



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